

Lab 3 – Architectural Selection Justification

Self-Service Coffee Kiosk

Architecture Selection: We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Choice: The system is divided into presentation, business, and data responsibilities. The Touch Screen Interface handles customer interaction, the Order Manager and Payment Service handle application logic, and the Menu & Pricing Database stores menu and pricing information.

Reason 1: Layered Architecture provides clear separation of concerns. The touch-screen interface can handle customer interaction without directly accessing the database, while the Order Manager manages the ordering process.

Reason 2: The architecture is suitable for a traditional kiosk application because the system has well-defined functions such as selecting coffee, processing payment, retrieving prices, and printing receipts.

Security Advantage: The layered structure can restrict direct access to the Menu & Pricing Database. Customer-facing components can communicate through the application layer instead of accessing stored data directly.

Performance Benefit: The architecture keeps frequently used operations such as menu retrieval and order processing organized within dedicated components, reducing unnecessary interaction between unrelated parts of the system.